About Me — Soumyadeep Nath

A living reference document for future projects, collaborators, and AI assistants. This version merges the prior profile with the new instructions, workflow rules, learning preferences, project context, communication standards, and the expanded career-automation mission.

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1) Personal / Professional Profile

Name: Soumyadeep Nath
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Current education: Executive PG Programme in Data Science & AI, IIT Roorkee
Academic background: M.A. and B.A. (Honours) in English Literature, Presidency University, Kolkata

I am transitioning deliberately from English Literature into Data Science, AI, and eventually Forward Deployed Engineer-style work. That humanities-to-data pivot is a core part of my identity, not something to hide. The value of the background is the combination of strong reading, writing, interpretation, and communication skills with quantitative, technical problem-solving.

The most accurate framing is:

> Literature taught me to think clearly and communicate well. Data Science gives me analytical and technical leverage. Together, they let me build systems and explain them.

Positioning principles - Be honest about skill level and scope. - Prefer accurate framing over inflated titles. - Undersell rather than oversell. - Treat credibility as a long-term asset.

Current role framing - Aspiring Data Scientist - AI / ML learner and builder - Full-stack AI product builder - Long-term target: AI Engineer -> Forward Deployed Engineer

2) Communication & Writing Style

I prefer responses that are precise, structured, and explanatory in the style of a world-class expert teacher. I do not want vague reassurance or shallow summaries. I want the reasoning, the trade-offs, and the mental model.

Tone - Professional, authoritative, analytical - Clear and pedagogical - Formal without sounding robotic - Confident, but never exaggerated - Direct when something is weak or unclear

Preferred formatting - Clear headings and subheadings - Dense, useful prose for reasoning - Tables for structured comparisons - Step-by-step breakdowns for complex ideas - Phase-wise plans, roadmaps, architectures, and evaluation frameworks - LaTeX or schema-style notation when it improves clarity

Words and habits to avoid - Casual filler - Hype language - Unverified claims - Hand-waving explanations - Overly simplified answers that skip logic - Generic motivational padding

Writing standards I value - Information density over fluff - Accuracy over speed - Explanation of why something matters, not only what it is - Iterative refinement: draft -> critique -> improve -> tighten

Teaching style I respond best to When explaining a concept, use: 1. Concept 2. Breakdown 3. Analogy / example 4. Visual enhancement when useful 5. Why it matters 6. Mini check

For larger lessons, finish with: 1. Final Summary 2. Key Takeaways 3. Knowledge Check Quiz 4. Answer Key 5. Your Next Step - Personalized Feedback

Coding help preference - Give a problem clearly first. - Offer a hint or skeleton before the full solution. - Do not reveal the final answer before I attempt it. - After I try, evaluate carefully and explain what was right, what was wrong, and why. - Keep the learning loop iterative.

3) Goals & Current Direction

Career goals - Land an entry-level role in Data Science, AI, or ML - Build toward AI Engineer and FDE-style roles over time - Strengthen Python, SQL, statistics, machine learning, and R - Build a portfolio that proves execution, not just learning - Use projects to create employability and visibility

Role targets - AI / ML internships - Entry-level Data Science roles - Generative AI roles - RAG / AI product engineering roles - Long-term: systems-heavy, customer-facing, deployment-oriented roles

Time and learning constraints - Limited daily time - Budget sensitive - Prefers free or low-cost resources - Wants the fastest path that still builds real skill

Current horizon - Finish the altitude/heat FIFA World Cup sports analytics project - Build the companion Next.js site for that project - Improve GitHub profile metadata and portfolio presentation - Keep progressing in Python, SQL, R, and ML

AI-Native Job Agent Architecture (Ongoing Mission)

A major long-term initiative is the construction of an AI-native autonomous job search ecosystem designed to automate and orchestrate the full employment acquisition lifecycle. Rather than functioning as a single application, the system is conceived as a network of specialized agents operating under centralized coordination and governed by a shared structural data layer.

The architecture is built around a **Candidate Profile JSON** that serves as the canonical source of truth. Every subsystem reads from and writes to this shared profile, ensuring consistency across discovery, application, learning, and orchestration workflows.

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DATA LAYER

 [10] Candidate Profile JSON
        Universal source of truth consumed by all agents

DISCOVERY

 [3] Gleaner
        Discovers and aggregates relevant job opportunities
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INTELLIGENCE

  [4] Research Agent
        Performs company and role intelligence gathering

  [5] Future Fit
        Monitors labor-market trends and emerging skill demand

APPLICATION

  [1] AlignResume
        Tailors resumes to specific job requirements

  [2] Overture Outreach
        Generates and delivers targeted cold-email campaigns

  [7] PDF Auto-Apply Agent
        Automates application form completion and submission

LEARNING

  [8] Memory Module
        Tracks application history, outcomes, and interactions

  [9] Sentiment Classifier
        Interprets recruiter responses and engagement signals

COORDINATION

  [6] Conductor
        Orchestrates workflow execution across all agents
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The strategic objective is to progressively integrate existing portfolio projects—**AlignResume, Overture, Gleaner, and Future Fit**—into a unified agentic platform capable of autonomously discovering opportunities, conducting market research, tailoring application materials, executing outreach campaigns, tracking outcomes, and continuously improving through feedback loops.

Beyond its practical value as a personal career-acceleration system, this initiative serves as a long-term demonstration of capabilities in **agent orchestration, workflow automation, structured data systems, LLM integration, decision intelligence, and Forward Deployed Engineering-style problem solving.**

4) Projects & Portfolio

I prefer projects that are real, high-signal, end-to-end, and employability-relevant. I value projects that combine data engineering, modeling, deployment, and documentation.

AlignResume An AI-powered resume optimization platform built with Next.js, TypeScript, and the Groq API. It focuses on resume-to-job-description matching, gap analysis, bullet rewriting, and tailored resume generation. A major design principle is factual grounding: the system should not invent credentials or fabricate experience.

Key themes: - Truthfulness guardrails - Structured JSON pipelines - Match scoring and explainability - Side-by-side resume comparison - Polished proof artifact / PDF output

Future Fit A skill trend intelligence platform built from a large real-world job-postings dataset. It analyzes skill demand trends over time, co-occurrence patterns, and emerging market signals.

Core ideas: - Time-series analysis - Market basket / Apriori association mining - Forecasting skill demand - Interactive analytics dashboard - LLM-assisted skill gap guidance

Altitude & Heat Adaptation in FIFA World Cups An ongoing sports analytics research project studying whether altitude, heat, humidity, and travel fatigue affect football performance across World Cups.

Core ideas: - Real-world sports data - Environmental and geospatial features - Statistical inference in R - GLMM / Negative Binomial modeling - Range-based projections rather than overconfident point forecasts - Formal hypotheses, architecture planning, risk registers, and evaluation structure

Gleaner A multi-board job scraping and ETL pipeline. It collects, normalizes, deduplicates, and routes listings from multiple platforms.

Key ideas: - Adapter-pattern architecture - Multi-source ingestion - YAML-driven workflow control - Google Sheets sync - Practical workflow automation

Overture A cold email writing and sending automation tool. It demonstrates production hardening, test coverage, and disciplined engineering.

Key ideas: - Deterministic formatting plus LLM polishing - Secure Gmail / OAuth flow - Async execution - Telemetry and metrics - Frontend dashboard for visibility

Portfolio principles - Show concrete outcomes - Favor deployed work - Prefer real data over toy examples - Build systems, not isolated demos - Make the work easy to inspect and understand from the README

5) Technical Profile

Current skill snapshot - Python: basic to early practical level - SQL: beginner - Statistics: descriptive statistics and foundational inference - R: actively learning; important for statistical modeling and reproducible analysis - Machine

Learning: foundational learning in progress - GenAI / LLMs: practical interest, especially structured prompts, guardrails, and workflow use cases

Languages and frameworks - R - Python - TypeScript - Next.js - HTML / CSS / JavaScript - Streamlit - Plotly

Data, ML, and analytics tooling - Groq API - Prophet - Apriori / association rules - Pandas and NumPy - StatsBomb and FBref data sourcing - OpenStreetMap, Open-Elevation API, Meteostat API - Firecrawl API - Google Sheets API

Engineering and DevOps tooling - Git and GitHub - GitHub Actions - Docker - pytest - Redis - Prometheus - ReportLab

Deployment - Vercel - GitHub Pages

Current learning resources - Python for Data Analysis by Wes McKinney - NumPy cheat sheet - Pandas cheat sheet - Python beginner cheat sheet - Google Colab / Jupyter Notebook

Technical values - Reproducibility matters - Clean pipelines matter - Testing matters - Strong naming matters - Modular design matters - Documentation matters

6) Workflow Preferences

Spec-Driven Development I prefer Spec-Driven Development. The spec comes first; code comes second.

The preferred sequence is: 1. Problem definition 2. Requirements 3. Architecture 4. Implementation 5. Evaluation 6. Hardening

Operating principles - No code before a spec exists - Documentation should explain why, not just what - Root-cause analysis before fixes - Iterative delivery is normal - Planning should be explicit - Edge cases should be surfaced early - Use test suites to prevent regressions - Balance spec detail with execution cost

How I like collaborators / AI assistants to behave - Act like a senior advisor, architect, researcher, and instructor - Verify design decisions against the spec before coding - Be direct when something is weak - Prefer evidence-based reasoning - Keep the work useful for employability - Treat technical work as a system, not a single answer

Debugging preference - Identify the actual cause first - Explain it clearly - Fix the cause, not just the symptom - Update tests if needed - Avoid quick patching or surface-level band-aids

Decision-making style - Think in trade-offs - Consider second-order effects - Ask what the choice means for portfolio, credibility, and long-term trajectory - Prefer evidence, comparisons, and measurable criteria

7) Learning Style

I learn best through structured breakdowns, examples, guided practice, and reflection after building.

What works best - Intuition first - Step-by-step mechanics - Worked examples - Hints and scaffolds before full solutions - Project-based learning - Clear feedback after an attempt

What I do not want - Answers before I try - Unstructured explanations - Abstract theory without application - Skipped steps - Overly terse guidance

Coding problem format I respond to best 1. Clear problem statement 2. Constraints and examples 3. Attempt it myself 4. Receive targeted feedback 5. Improve and iterate

Adaptive learning mindset I respond well to increasing difficulty, structured evaluation, and visible progress tracking.

8) Explicit Explanation Contract

When explaining concepts to me, the default should be a very thorough teaching format.

Required structure - Concept - Breakdown - Analogy / Example - Visual Enhancement - Why it matters - Mini Check

Required closing sequence 1. Final Summary 2. Key Takeaways 3. Knowledge Check Quiz 4. Answer Key 5. Your Next Step - Personalized Feedback

Non-negotiable teaching style - Do not assume prior knowledge - Do not stop at the surface level - Use simple words where possible - Make the reasoning visible - Include everyday examples - Include a visual whenever it helps

9) Design & Aesthetic Preferences

- Interactive dashboards over static charts
- Dark-mode / sports-analytics aesthetic
- Clean, modern, slightly premium feel
- Strong typography and hierarchy
- Consistent design tokens
- Navy blue and gold works well in branding contexts

What I usually want improved - GitHub profile README - Project descriptions - Portfolio about sections - LinkedIn-style positioning - Release announcements - README summaries - Hero copy and slogans

10) Resume, Job Search, and Visibility

Resume principles - Honest, evidence-based, and version-controlled - Programmatic generation is preferred over manual editing - Every listed skill should trace to a real project or documented evidence - No inflated claims - Put the strongest proof early

Job search strategy - Best-fit areas often include EdTech, IT services / consulting, and media / content-tech - Roles involving GenAI, RAG, data analysis, and AI product work are particularly relevant - Fresher-friendly search terms matter

Visibility goals - Improve repo metadata - Keep public projects polished - Make the portfolio easy to inspect - Use deployed work as proof of capability

11) Behavioral Patterns I Seem to Exhibit

These are recurring patterns I should expect myself to keep showing.

Career strategy - I treat the transition into data science as a deliberate reinvention - I prefer portfolio-first credibility - I think long term rather than chasing only immediate wins

Project selection - I prefer high-signal projects - I prefer real-world problems - I prefer end-to-end ownership

Learning and decision-making - I structure before I act - I care about second-order effects - I prefer evidence to opinion - I like systems thinking - I want deep understanding, not surface memorization

Communication - I prefer precision over brevity - I like executive-level organization - I improve work through iterative refinement - I care about credibility and differentiation

12) Living Notes

This document should evolve as I: - complete new projects - improve technical skills - refine my portfolio - change career targets - update my workflow preferences

Best use cases for this document - New project onboarding - Resume and LinkedIn writing - Portfolio copy - AI assistant context - Job search support - Technical tutoring - Long-form planning

Short onboarding summary for new projects If a new collaborator only needs the minimum context, the core facts are: - I am Soumyadeep Nath from Kolkata, India. - I come from English Literature and am transitioning into Data Science and AI. - I value honesty, structure, and precision. - I prefer spec-first, root-cause-first work. - I want explainers that are clear, deep, and well organized. - I am building a serious portfolio through real projects.

13) Update Log

- Added explicit teaching and quiz format preferences
- Added adaptive coding / hint-first learning preferences
- Added stronger workflow rules around specs, debugging, and evaluation
- Added project and portfolio context from recent work
- Added career direction: DS -> AI Engineer -> FDE
- Added AI-Native Job Agent Architecture as an ongoing mission

- Consolidated communication, design, learning, and job-search preferences into one living reference

This document is intended to be pasted into future projects so I do not need to re-explain myself from scratch.